EXPERIMENT 10

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

HTML

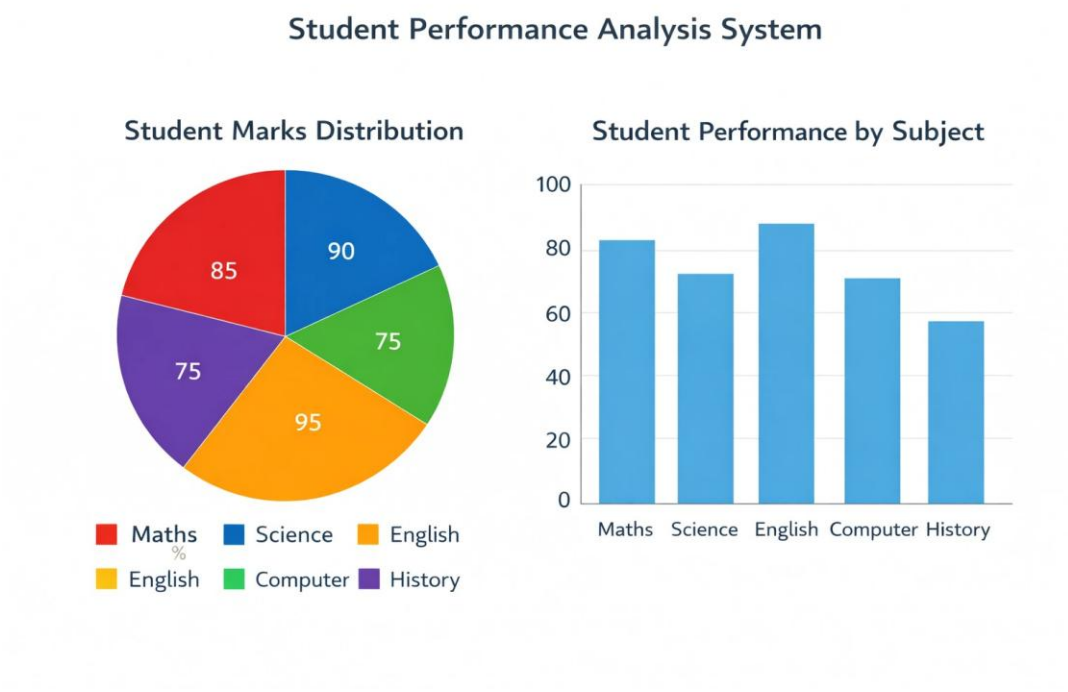
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Student Performance Analysis System</title>

    <!-- Chart.js -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <style>
        body {
            font-family: Arial;
            text-align: center;
            margin: 50px;
        }
        canvas {
            margin: 20px;
        }
    </style>
</head>

<body>

<h2>Student Performance Analysis System</h2>

<canvas id="pieChart"></canvas>
<canvas id="barChart"></canvas>

<script src="script.js"></script>

</body>
</html>
```

STEP 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```javascript
var labels = ["Maths", "Science", "English", "Computer", "History"];
var dataValues = [85, 90, 75, 95, 70];

var pie = document.getElementById("pieChart").getContext("2d");

new Chart(pie, {
    type: "pie",
    data: {
        labels: labels,
        datasets: [{
            label: "Marks",
            data: dataValues,
            backgroundColor: ["red", "blue", "green", "orange", "purple"]
        }]
    }
});

var bar = document.getElementById("barChart").getContext("2d");

new Chart(bar, {
    type: "bar",
    data: {
        labels: labels,
        datasets: [{
            label: "Marks",
            data: dataValues,
            backgroundColor: "skyblue"
        }]
    },
    options: {
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});
```

OUTPUT:


Student Performance Analysis System
Student Marks Distribution
85
90
75
75
95
Maths
Science
English
English
Computer
History
Student Performance by Subject
100
80
60
40
20
0
Maths
Science
English
Computer
History